Assembly of a Tentacle Robot - Robotic Arm for Asteroid Mining/Space Exploration

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ABSTRACT

This project focuses on the design, assembly and evaluation of tentacle robotic arm made for asteroid mining and space exploration. The primary objective is to develop a flexible and adaptable robotic arm capable of manipulating objects in dynamic and unstructured environments, mimicking the movements of biological appendages like tentacle or snake-like structures. The significance of this research lies in its potential to enhance robotic capabilities in space, particularly in tasks that require precision and adaptability, such as asteroid mining.

To achieve the project objectives, a structured approach was used, starting with a comprehensive assessment of the available materials and hardware, followed by the assembly of the robot using inherited and newly sourced components obtained within the project's budget. The control system was developed using the Dynamixel Wizard UI and a custom Python script, allowing for manual control and a potential approach for semi-automated control of the robotic arm. Two test cases were conducted: an object interaction test to demonstrate the robot's agility and precision, and a coil formation test to display the robot's flexibility and ability to perform complex manipulations.

The success of these tests confirms the robot's potential for space applications, offering a foundation for future development and research. Future work may explore more advanced control algorithms, the use of alternative materials for enhanced durability and mobility, and improve automation using machine learning techniques, allowing for more advanced and capable robotic systems in space exploration. This dissertation provides a detailed analysis of the design process, challenges encountered and the innovative solutions implemented, contributing useful insights to the field of robotics in space.

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1 INTRODUCTION

The field of robotics has impacted plenty of industries as technology has continued to advance. One of these industries would be space exploration. In its nature, space exploration is a highly dangerous endeavour and only a number of people have been fortunate enough to be able to go up to the International Space Station and even fewer have been able to go further than that [1]. Given the limitations human beings have in space, we rely on the use of machines for exploration and acquiring substances from space. As such that is the main scope of this project.

Currently, the International Space Station (ISS) has two robotic arms: the Canadaarm2 and the Japanese Experiment Module Remote Manipulator System (JEMRMS). The goal of this project is to create an arm similar to these arms that can be used to grasp objects, primarily asteroids. The robot will grasp objects in a tentacle or snake-like manner, coiling around its target.

1.1 Background and Context

The field of robotics has impacted numerous industries as the underlying technology for them has continued to evolve. One of these industries in which it plays a crucial role is space exploration. The exploration of space presents unique challenges which demand rigorous training to combat them for astronauts and innovative technological solutions. One of the solutions is the implementation of robotic systems which are capable of handling diverse, unpredictable and dangerous environments. Several existing robotic technologies and missions have been deployed and are still functioning at the moment. For example, the National Aeronautics and Space Administration's (NASA) Robonaut (and its following versions), was designed for agile manipulation in microgravity situations [2]. Similarly, the European Space Agency's (ESA) robotic arm was designed for detailed sample collection and analysis [3]. Both these projects provide insight into the challenges of robotic mobility in space and the importance of reliable robotic systems for planetary exploration. They also highlight several examples of the foundation of the concept of this project, the development of a tentacle robot arm.

The concept of a tentacle robotic arm draws inspiration from biological ecosystems, mainly the flexible appendages of cephalopods i.e. octopuses. These animals are known for their ability to gather objects with precision and adaptability, abilities ideal for any robotic system used in space exploration. Therefore, the robotic arm aims to replicate these abilities, creating a system that can navigate irregular surfaces such as asteroids, and grasp and control objects of different sizes.

Following up on the robotic arm's ability to grasp objects of irregular sizes, a concept that would be discussed is to demonstrate the arm's potential applications in asteroid mining (which was briefly mentioned above). Asteroid mining is a revolutionary approach to resource extraction (in space), with its primary aim to extract valuable materials from asteroids in space. Initially, the feasibility of this concept was rather low, but given the incredible improvements in space technology, robotics and artificial intelligence, which as a group allow the identification and processing of the extracted materials in the challenging space environment, the possibility of asteroid mining has never been more plausible. For example, NASA successfully collected samples from asteroids using their Touch-and-Go Sample Acquisition Mechanism (TAGSAM) in their OSIRIS-Rex mission [4].

1.2 Scope and Objectives

The scope of this project is focused primarily on the assembly and control of the robotic arm.

Project Objectives:

- Inventory Assessment: Identify and catalogue all available motors and robot arm segments.
- Design Phase: Explore different systems to successfully create a visualisation of the tentacle robot arm fully assembled.
- Control System Development: Develop a preliminary control system framework capable of basic movement operations.
- Prototype Assembly: Assemble the physical tentacle robot arm prototype using the available cables, motors and arm segments.
- Testing & Iteration: Conduct testing with the assembled prototype to identify any mechanical or control issues.

1.3 Achievements

The tasks mentioned above have all been completed within the duration of this project. A detailed assessment was carried out on all the inventory equipment provided. Utilising the available inventory, a tentacle arm manipulator was successfully assembled and mounted simulating its ability to manipulate objects with a control system controlled by a user. Various tasks were conducted to test the limits of the robot to mimic its potential in space. One of the significant challenges encountered involved creating a zero-gravity simulation environment to fully assess the robot's full capabilities in spatial manipulation and potential applications in asteroid mining. To address this challenge, several innovative approaches were explored, which are outlined later on in this report.

1.4 Overview of Dissertation

Henceforth, this dissertation shows the extensive background research and technical work conducted within the project timeline to achieve the outlined objectives above. The background research will focus on the robotic applications in space, the advancements in robotics for spatial manipulation and the mechanisms of tentacle (continuum) robots, including both existing and future designs. Given the project's primary focus on the assembly and control of a robotic system, the technical report will detail the software tools used to facilitate tentacle robot's movement, as well a detailed explanation of the hardware components utilised. The challenges and limitations encountered due to the project's scope and budget will also be examined. Subsequently, the final assembly will be presented through visual evidence, displaying the robot's motion and manipulation capabilities. Finally, recommendations for future improvements to the project will be provided.

2 BACKGROUND THEORY AND LITERATURE REVIEW

2.1 Early Space Missions

As discussed earlier in this document, the primary focus of this project is space exploration. Robots have played an integral role in space missions from the very beginning, largely due to the inherent risks associated with the harsh environment of space. The first successful launch was conducted by the Soviet Union on October 4th, 1957. This mission was the first artificial satellite, Sputnik 1, to be sent into space, inaugurating the space race [5].

It was not until four years later, on April 12th, 1961, that a human-occupied spacecraft successfully reached space. This mission, known as Vostok 1, was piloted by the Russian astronaut Lt. Yuri Gagarin, who became the first human to orbit the Earth [6]. The risks involved in this mission were unprecedented, especially considering the technological limitations of that era. This mission's success was driven by the intense competition of the ongoing space race.

Eight years later, on July 20th, 1969, the famous Apollo 11 mission-also known as the Moon Landing-marked a monumental milestone in space exploration when humans set foot on the Moon. This mission, led by NASA, was a significant achievement that captured the imagination of people worldwide and emphasized the rapid advancements in space technology ensuring mission success during that period.

The early missions highlighted above provided the foundational groundwork for the field of space exploration, particularly in the development and application of robotic systems. The success of missions like Apollo 11 spurred the advancement of robotic technologies, which evolved from basic tools and instruments to more sophisticated systems, including robotic arms and manipulators.

This period of rapid technological progress directly influenced the development of robotic arms, such as the Canadarm project (discussed later on in this report), which became an essential component of space missions. The continued evolution of robotic systems has now led to the conceptualization and development of more advanced robotic systems, like the tentacle robot arm in this project, which offers enhanced flexibility and functionality in space environments. The lessons learned from these early missions, combined with the advancements in robotics, have paved the way for the assembly and deployment of such innovative systems in modern space exploration.

2.2 Advancement of Robotics in Space Exploration

As the foundational groundwork for space exploration was established, more technological innovations and space deployments quickly followed, which aimed at achieving more ambitious goals in future missions. Robots have been crucial to these innovations, significantly improving the success rates of exploration missions. Initially, robots on missions focused on tasks requiring high precision, which were impractical or dangerous for human astronauts. One of the earliest examples of this was the use of the robotic arms on the Apollo Lunar Modules, which were designed to collect soil samples and deploy scientific instruments. Although these systems were relatively simple and manually controlled, they set the stage for more advanced robotic technologies, including those that have inspired the design of tentacle robots like this project's focus.

Still a leader in the early space race, the Soviet Union launched Lunokhod 1 on Novemeber 10th 1970, which became the first robotic rover to operate on another planetary body. Lunokhod 1 was a remote-controlled vehicle equipped with cameras, instruments and sensors to measure the Moon's physical and chemical properties [7]. Despite facing challenges such as extreme cold (challenges and limitation will be explored later on), Lunokhod 1 showed the potential of robotic systems in exploring harsh and remote environments. This insight was essential for developing more adaptable robots such as tentacle robots or any other form of manipulator for future missions.

In the 1970s, NASA began development of a new space transportation system, the Space Shuttle. This project highlighted the need for advanced robotics to perform complex tasks in space, such as unloading equipment/payloads from the shuttle's bay [8]. This brought about the creation of the Canadarm, a robotic arm designed specifically for the Space Shuttle. Designed to be operated by astronauts onboard the shuttle, Canadarm displayed the effectiveness of teleoperation (control from a distance) under real-time constraints. The Canadarm's design and success have provided underlying information for the development of more advanced, flexible robotic systems, such as the tentacle robot, which can manipulate objects in dangerous environments.

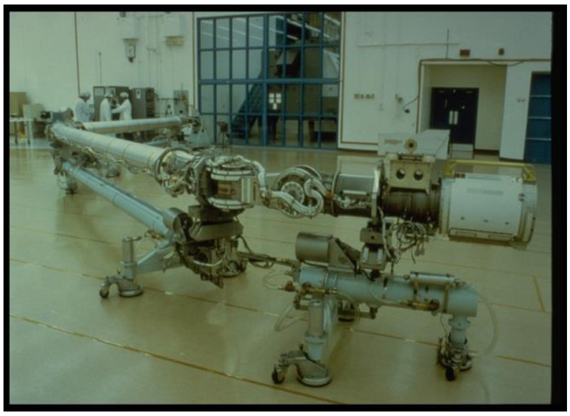

Figure 1: Canadaarm [8]

As missions strayed further from Earth, the delay in communication (up to approximately 20 minutes one way to Mars) required robots capable of autonomous operation making independent decisions without real-time input from Earth [9]. This led to the development of more advanced robotic systems. NASA's Mars Pathfinder mission required this shift, with the Sojourner rover becoming one of the first to operate with remarkable autonomy. Launched on December 4th, 1996, Sojourner's primary objective was to navigate the Martian surface and perform scientific experiments with minimal input while avoiding obstacles [10]. The success of Sojourner (which led to the subsequent rovers like the Spirit and Opportunity) emphasised the importance of autonomy, flexibility and adaptability. These key characteristics are aimed to be explored during the development of the tentacle robot [10].

The development from simple, rigid robotic arms (much like the arm in this project), to more advanced autonomous systems provide an important foundation for developing tentacle robots, which provide increased flexibility and capability. These capabilities are particularly important for complex space missions, such as asteroid mining or deep space exploration, where robots would need to navigate uneven surfaces, manipulate delicate objects or manoeuvre in tight spaces. The lessons learned from earlier robotic systems such as the Canadarm or the Mars rover helped to inform the design and assembly of tentacle robots, which is the focus of this project.

2.3 Continuum Robots and Robotic Arms

Continuum robots, also known as tentacle robots, are a significant deviation from traditional rigid robotic manipulators. Unlike rigid robotic structures, continuum robots are characterised by their flexible and continuous body structures mimicking cephalopods, meaning that they can bend, twist and elongate in a manner similar to biological organisms such as octopuses' arms, elephant trunks and snake-like bodies [11]. The advantage of these robots is that they are highly adaptable and capable of dexterous movements, making them more ideal than their rigid partners.

Continuum robots are described by their ability to perform, smooth continuous movements along their full length, which will allow them to navigate and manipulate objects in environments that are complex and unstructured. This flexibility is achieved through a series of segments that can bend independently, allowing movements in multiple directions without needing discrete joints used in conventional robots. Ideally, the absence of rigid joints gives continuum robots an infinite number of degrees of freedom, which makes them highly capable of performing a wide range of tasks that may require manipulation [11].

With the design of tentacle robots, some several key principles or characteristics must be taken into account are:

- **Material**: Continuum robots are normally constructed from soft and flexible materials such as silicone or rubber. With these materials, the robot is able to bend and stretch without breaking, mimicking the properties of biological tissues [12]. In some other designs, flexible metal or some forms of composite materials are used to provide a balance between flexibility and strength, which then allows the robot to apply more strength. Later on in this report (in the technical chapters), the materials used in this project will be elaborated on.
- **Control Mechanism**: The control system for continuum robots is inherently different from that of rigid robots because of their continuous nature. Conventional robots use motors and actuators at individual joints, while continuum robots mainly use pneumatic, wire-driven or hydraulic systems to achieve movement. For example, wire-driven systems use cables which would run along the length of the robot, contracting and relaxing to create various movements.
- **Types of Movement**: The advantage of continuum robots is they can perform a vast range of movements such as bending, twisting, elongating and compressing [13]. A combination of these movements is generally used to navigate and accomplish complex and delicate tasks that may require high levels of precision.

The unique properties of continuum robots make them well-suited for use in particular environments where rigid robots would not suffice. Although the primary focus of this project is exploring their applications mainly in space exploration, it is important to highlight some of their other applications in different mediums. In the medical world, continuum robots such as the STIFF-FLOP (STIFFness controllable Flexible and Learnable Manipulator for Surgical Operations) robot have been used in minimally invasive surgeries. It was developed to assist surgeons in accessing hard-to-reach pathways and navigating through narrow pathways in the human body [14]. With a robot using soft, flexible material, the risk of tissue damage is also greatly reduced. In underwater explorations, continuum robots are used as manipulators when inspecting and maintaining underwater structures such as pipelines and oil rigs [15], somewhat mirroring the use of these robots in space. In industry/manufacturing automation, continuum robots' ability to manoeuvre around corners or into tight spots makes them ideal for assembling delicate components that might require precision and flexibility.

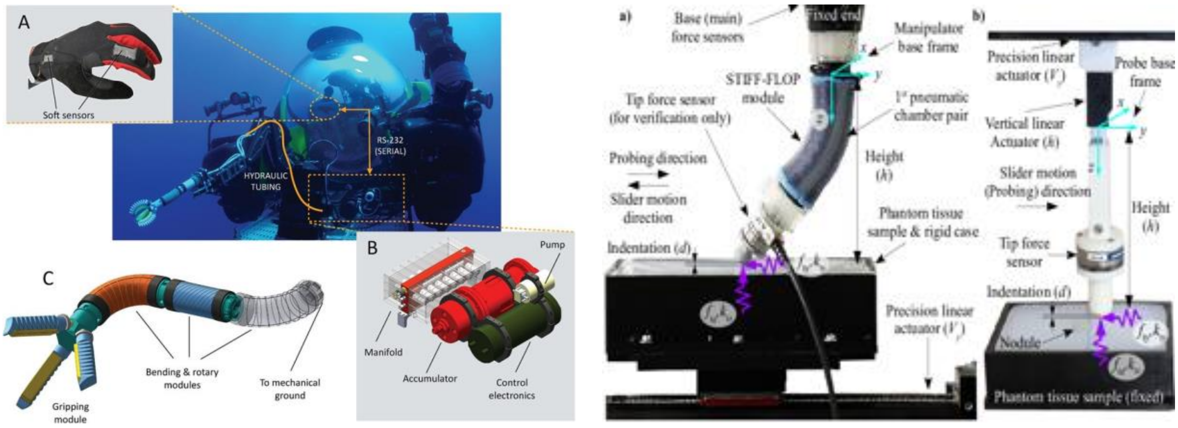

Figure 2: STIFF-FLOP Robot and Underwater Continuum Robot example [16], [17]

2.4 Potential in Space Exploration

As the main environment this project concerns, the application of continuum robots in space exploration and utilization is an exciting potential. Their unique capabilities could help address some of the key challenges of operating in the harsh and unpredictable environment of space.

A potential application in space is in Asteroid Mining. Asteroid Mining refers to the extraction of valuable materials from small celestial bodies. The main motivation for asteroid mining is to collect resources that are rare or no longer available on Earth such as precious materials e.g. platinum. Asteroids, particularly those that are considered Near-Earth Objects (NEOs), are known to be substantial sources for these materials. The feasibility of asteroid mining has amassed significant interest in recent years, especially from space agencies like NASA. Below are brief descriptions on existing robotic continuum robot concepts that have been developed in industry that have potential in asteroid mining and space exploration:

- **Asteroid Redirect Mission**: NASA planned the now cancelled Asteroid Redirect Mission (ARM), which was a robotic system that was designed to capture and manipulate small asteroids. Among its features were multiple tentacle-like robotic arms equipped with grippers capable of enveloping and successfully grabbing the surface of the asteroid [18].

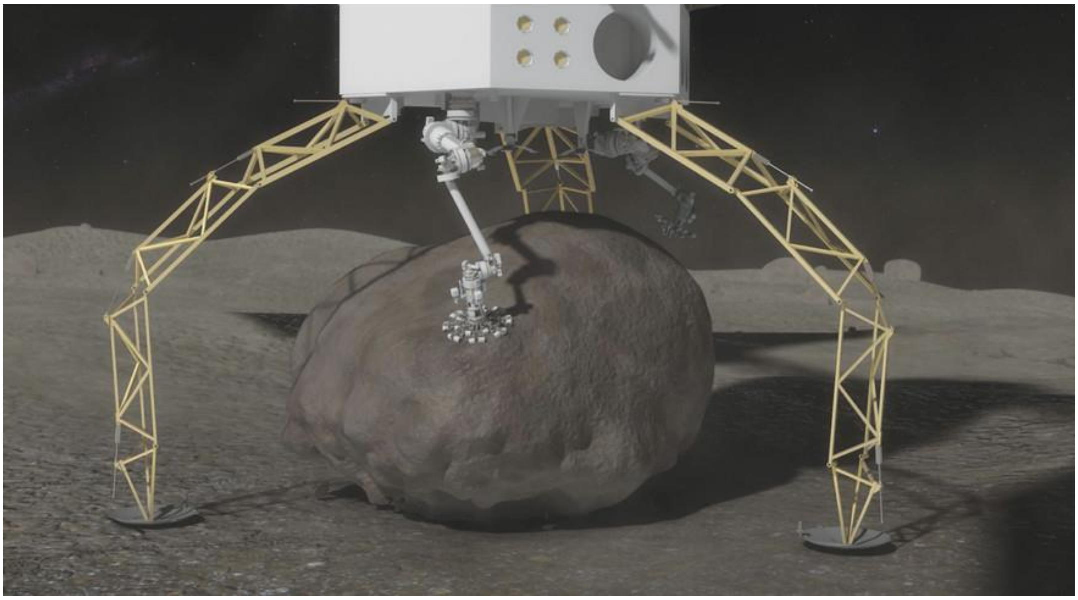

Figure 3: Concept of ARM [19]

- **PRO-ACT**: The Planetary Robotic Actions for Compound Tasks. This project, funded by the European Space Agency, involves developing a robot that is capable of performing tasks in space such as assembling structure or making repairs. The project continues exploration of the use of tentacle-like robots for these operations due to their flexibility and adaptability.

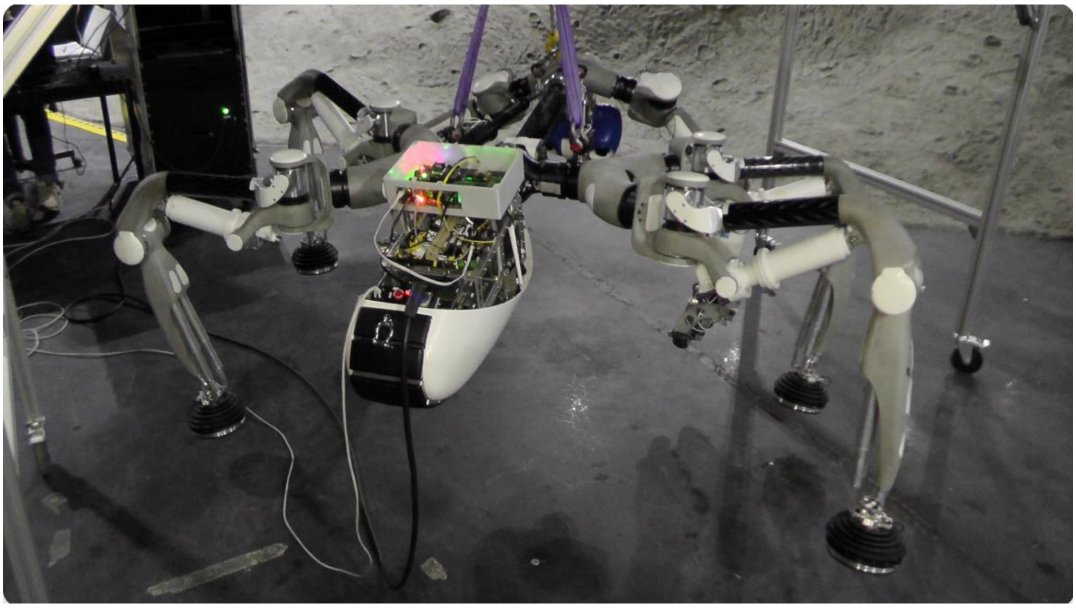

Figure 4: PRO-ACT[20]

- **RoboSimian**: The RoboSimian is a four-limbed robot designed by NASA's Jet Propulsion Laboratory (JPL) created to operate in disaster response scenarios, but its design principles are also applicable to space exploration. It utilises a flexible limb structure to navigate and manipulate objects in its environment [21]. The multi-functionality of the RoboSimian's limbs somewhat mimic the capabilities of continuum robots, allowing it to be a potential application to asteroid mining.

Figure 5: RoboSimian [21]

- **OSIRIS-REx Sampling Arm**: This NASA mission was designed to collect samples from a specific asteroid that was being observed, asteroid Bennu, which was accomplished successfully. The spacecraft sent in this mission was equipped with the Touch-And-Go Sample Acquisition Mechanism (TAGSAM). This was a robotic arm with a flexible sampling head that could make contact with Bennu's surface, collect the material and return to the spacecraft [4]. While the craft does not classify as a continuum robot, TAGSAM arm's ability to reach out, touch and manipulate the asteroid's surface highlights the need for robots that use both flexibility and precision when in motion in space. Conceptually, tentacle robots could improve the robot's capability by offering more adaptability.

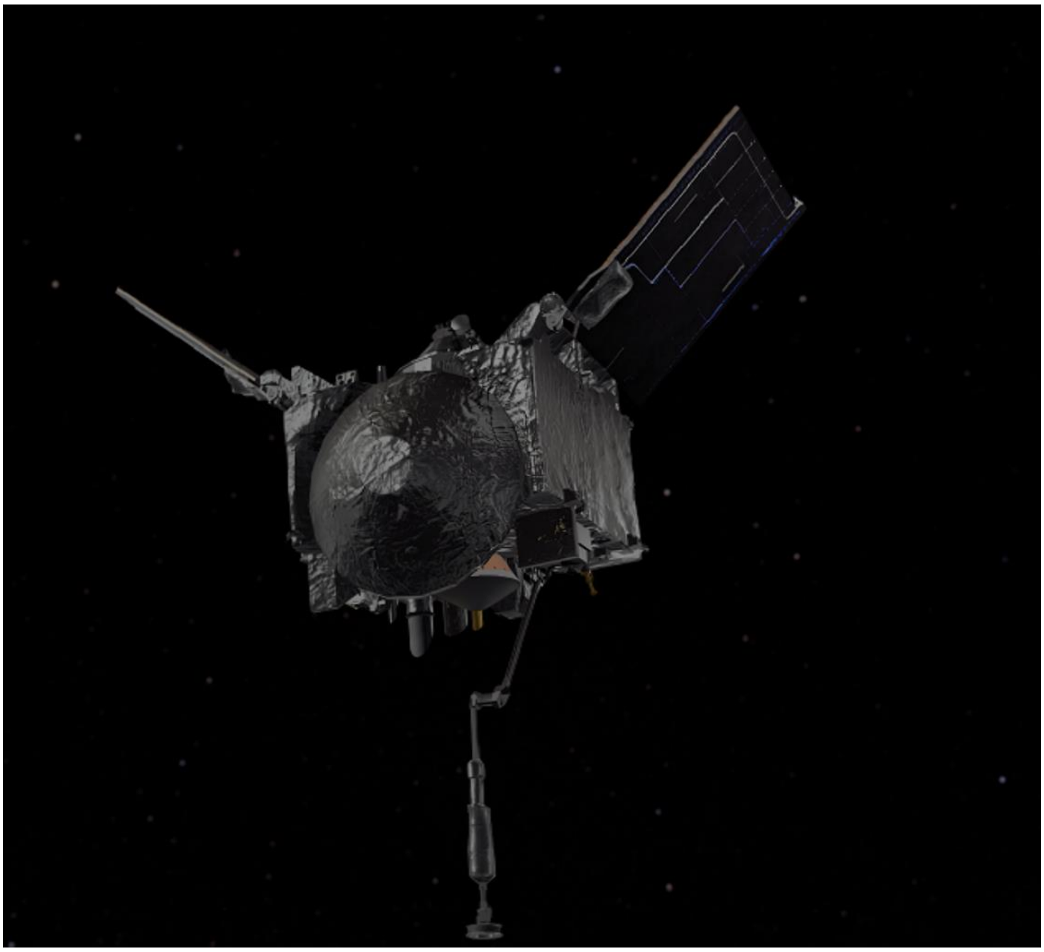

Figure 6: OSIRIS-REx[4]

While continuum robots, such as the few mentioned above, present significant advantages in flexibility and adaptability, it is important to compare them with rigid robotic arms previously and currently being used to better understand the benefits they provide. For example, the Canadarm (Section 2.2) and its successor, Canadaarm2, have been highly effective on the Space Shuttle and the International Space Station (ISS) for tasks requiring precision and load bearing. However, due to their rigid structures, they have limited ability to adapt to unstructured environments such as asteroid surfaces. Similarly, the European Robotic Arm and Japanese Experiment Module Remote Manipulator System (JEMRMS) succeed at performing precise maintenance and assembly for the ISS but lack the flexibility needed for more unpredictable environments [3], [22]. These limitations highlight the potential of continuum robots to improve robotic capabilities in space, especially in asteroid mining, where adaptability to unpredictable, changing terrain and the ability to manoeuvre around obstacles are important.

2.5 Challenges and Limitations of Robotics in Space

Robotics has become an essential tool in space exploration, allowing missions to explore farther and carry out tasks that would be impossible or too dangerous for human astronauts. However, the use of robotics in space is accompanied with numerous challenges due to the harsh and unpredictable conditions beyond the Earth's atmosphere and technological limitations.

Prior to detailing all the challenges, the greatest challenge all space craft leaving earth's atmosphere all have to address is the harsh environmental condition of space. Space is filled with high levels of radiation from solar and cosmic sources. Unlike Earth, which has an atmosphere and electric field that provide protection, in space, robots are exposed to damaging radiation. This leads to the degradation of useful components, sensors and eventually, structural material [23]. To combat this, radiation-hardened specifications are required thereby leading to increased mission costs and weight, which in the case of manipulators, reduces flexibility design capabilities.

Another factor regarding the environment, would be the temperatures in space. In space, temperature can vary widely ranging from extremely cold in the shadows of planets to extremely hot when in contact with direct sunlight. Fluctuation like this can cause materials used on robots to expand and contract leading to structural failures or loss of functionality.

Subsequently, robots deployed in space have to consider several other challenging factors before deployment:

- **Communication Delays and Autonomy**: The vast distance between Earth and exploration bodies like Mars result in communication delays. Due to this, robots operating at these vast distances must be capable of autonomous or semi-autonomous operation, making real-time decisions without any human control. This would require on-device AI and machine learning algorithms, which take time to develop and validate for unpredictable environments.
- **Limited Power and Resources**: Space missions typically rely on power from solar panels or nuclear power sources. Power management is crucial for the operation of any robotic system and any inefficiency might limit a robot's operational time and the range of tasks it can perform.
- **Maintenance and Repair Challenges**: Space missions have limited capacity for repair once a robot is deployed. This inability to perform scheduled maintenance to replace any faulty parts requires deployed robots to be exceedingly robust and reliable.

- **Inertia in Microgravity**: In microgravity environments, robots cannot depend on weight (or friction) to stabilise themselves. This makes tasks such as manipulating objects difficult, as the robot could push itself (or the object to be manipulated) away.

With the application of tentacle arm for asteroid mining, all the above challenges are included but with a significant focus on the arm's control complexity in microgravity. Depending on the set up of the tentacle arm, it could have an infinite number of degrees of freedom due to their flexible, continuous structure. The absence of a stable reference frame in microgravity means that every movement of the tentacle arm must be precisely controlled to avoid unwanted displacements. To circumvent this, advanced control algorithms and possibly real-time adjustments depending on feedback from multiple sensors located on the tentacle arm would become a necessity thereby increasing the computational load and power requirements for the robot [24] [25].

2.6 Summary

This chapter provided a comprehensive background and literature review, following the evolution of robotics in space exploration from its early use in fundamental missions, to the development of more advanced robotic systems today. For example, the transition from simple, rigid robotic manipulators, such as the Canadarm, to the more advanced flexible systems like continuum robots, showing the unique potential and advantages these tentacle robots offer for space applications. The literature above emphasized the critical role of robotics in tackling the adverse challenges of space environments, particularly for missions that require high adaptability and precision, such as asteroid mining.

Additionally, this chapter highlighted key challenges and limitations that accompany deploying robotics in space, including harsh environmental conditions, communication delays and the increasing need for autonomy. With the increased feasibility of asteroid mining, the technical complexities for developing a tentacle robotic arm for this task were discussed, highlighting the importance of advanced control algorithms and real-time adaptability in microgravity environments.

By showing the current state of research and technology, the above discussion sets the stage for the original contributions and designs made in the following chapters, where the technical design, development, and assembly of a tentacle robot for potential space applications are explored. The gaps identified, mainly in the areas of scale, budget, control complexity and environmental adaptability, show the need for unique solutions and provide the foundation for the solutions and designs suggested in this dissertation.

3 DESIGN AND DEVELOPMENT OF THE TENTACLE ROBOT

3.1 Robot's Objectives

The primary objective of this dissertation is to evaluate the tentacle robotic arm's effectiveness as a manipulator in grasping and interacting with objects. The project aims to demonstrate the robot's flexibility and adaptability, particularly in its ability to form complex shapes and configurations, such as a spiral, snake-like coil, which could be highly beneficial for applications like asteroid mining. The robot will be manually operated to test its overall performance in a simulated environment, highlighting its potential when scaled up for more demanding tasks in space exploration.

The scale and budget of the robotic systems detailed above are all an extraordinary scale. Given the scope of this research, the tentacle robotic arm in this project is created on a smaller scale but its base design will mimic rigid robots such as the Canadarm and additional actuators at joints will improve its adaptability creating a tentacle-like arm. The aim of the tentacle robotic arm after assembly, is to explore its ability as a manipulator in grasping and touching objects. To achieve this, the robot will be manually controlled by a user and the actuators installed with it must generate enough torque to overcome the forces of gravity, as the test carried out are simulated in a non-zero gravity area. Each of the actuators installed will increase the adaptability and flexibility of the robot, transforming it to a tentacle robot. Controlling the robot into a helix, spiral-like shape will also display the flexibility of the robot and its potential application (if built to a larger scale) in asteroid mining.

3.2 Robot Design

The design of the tentacle robotic arm draws inspiration from traditional rigid robots like the Canadarm but adds additional actuators at key joints to enhance adaptability and flexibility. This smaller-scale model prioritises dexterity, which is crucial for space applications. The actuators must be capable of generating sufficient torque (detailed later on) to counteract gravitational forces during testing in a standard gravity environment. By allowing the robot to achieve complex shapes, such as the spiral configuration, the design indicates its potential for more complicated manipulations crucial for tasks like asteroid mining in unpredictable environments.

3.2.1 Inventory Assessment

The materials and components used in this project were carried over from a previous continuum robot project. Consequently, the selection of materials and actuators was not made specifically for this project. Due to this, it was essential to maximise the utility of the materials listed below to ensure the success of the tentacle robot assembly:

BILL OF MATERIALS			
Component	Quantity	Total Price £ (Cost of acquiring item for project)	Source
Aluminium "arm" segments	11	0.00	Inherited from previous project
Aluminium "joint" segments	9	0.00	Inherited from previous project
Dynamixel MX-64 Actuator (with horn and bearing)	6	0.00	Inherited from previous project
Dynamixel MX-64 Actuator (without horn and bearing)	9	0.00	Inherited from previous project
FRO5-H101 (Foundation support attached to the ceiling)	1	0.0	Inherited from Previous project.
Dynamixel 3Pin Cable Pack 200mm (TTL communication) (10 pack)	2	31.60	Dynamixel Supplier
Dynamixel 3Pin Cable Pack 140mm (TTL communication) (10 pack)	2	28.40	Dynamixel Supplier
HN05-I101 Set (Horn	7	119.70	Dynamixel Supplier

set)			
HN05-N102 Set (Horn Set)	7	86.10	Dynamixel Supplier
M2.5 Screw Set	1	12.90	University Supplier
Rubber Band Set	1	0.00	University
HN05-I101 Set (Horn set) (3D Printed)	12	0.00	University
HN05-N102 Set (Horn Set)	12	0.00	University

Table 1: Bill of Materials Table

The initial budget given to this project was approximately £100 but it can be seen from the above costs that when working with robotic hardware, maintenance costs quickly increase (with the overall cost being approximately £278.70).

3.2.2 Hardware Challenges

As mentioned earlier, most of the inventory for this project was inherited from a previous project. This inheritance resulted in some faulty components being included, requiring additional part orders. A significant challenge was related the Dynamixel MX-64 actuators, which were crucial for this project's success. Although there initially appeared to be a sufficient number of actuators in the inventory (as shown in the table above), upon initial testing, several inherited units were found to be defective and unusable.

Due to budget limits and the high costs of these actuators, replacing the defective units was not feasible, forcing the project to be continued with the available components. This limitation had a direct impact on the length and consequently the adaptability of the tentacle robot. Further details regarding these challenges and their impact on the robot's performance will be highlighted in the actuation and kinematic sections.

3.3 Design Implementation

3.3.1 ROS

Robotic Operating System (ROS) is an open-source framework that provides the necessary tools and libraries to create robust robotic applications. It is designed to simplify the development of complex robotic behaviours across a wide range of robotic platforms [26]. In the assembly and simulation of the tentacle robot, ROS plays a vital role in enabling modular development and testing. In this project, two aspects of ROS are used to visualise the desired motion of the tentacle robot arm, URDF and RVIZ.

- Unified Robot Description Format: To simulate the tentacle robot, a detailed URDF file was created. The URDF is an XML format used in ROS to describe the physical configuration of the robot including its links and joints (and sensors, not applied in this project) [27]. In this project, the URDF file captures the tentacle robot's geometry, kinematics and dynamics, which enables approximate simulations in RViz. Please see the Appendix section for a link to the XML configuration.

- RViz (ROS Visualisation): This is a visualisation tool that allows the visualisation of the physical configuration URDF that was defined. Using RViz, the robot's kinematic configuration can be verified and different movements can be tested to explore the robots capabilities.

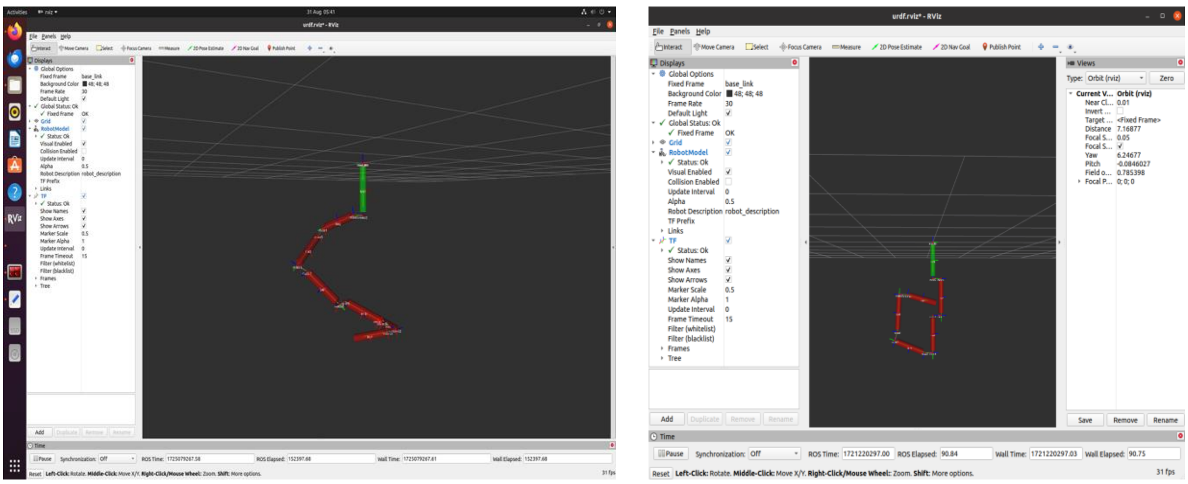

Figure 7: Visualisation of the Tentacle Robot Arm

3.3.2 Dynamixel MX-64

The Dynamixel MX-64 motors are a critical component in the design and functionality of the tentacle robot arm. Developed by Robotis, these motors have the capabilities for high precision, durability (see Figure 12 and Table 2) and advanced features, making them ideal for complex robotic applications such as the one presented in this project. The MX-64 motors provided the necessary torque, control accuracy, and flexibility needed for the robotic arm to accomplish its intended objectives (4.2.2).

3.3.2.1 Mechanical Challenges

Initially, integrating some of the MX-64 motors into the robotic arm posed certain challenges. The motors alone could not be securely attached to the arm structure without the use of additional components known as horns and bearing sets (Figure 8), which are required on either side of the motor facilitate rotation to specific positions (Figure 20). A significant issue arose because many of the motors inherited from previous projects did not include these horn sets, rendering them unusable in their original state (Figure 9). Although an order was placed for the missing horn sets, the delivery delays extended beyond the project timeline.

Figure 8: Horn and Bearing Sets for Motors[28]

Figure 9: Missing Horns and Bearing Sets.

To address this issue, a workaround was implemented using Computer-Aided Design (CAD) files to create 3D-printed replicas of the required horn components. While the original horns were made of metal, ensuring durability, the 3D-printed versions were made of plastic, presenting a potential risk of reduced durability. However, during assembly and initial testing, the plastic horns were sufficiently durable for the project's needs.

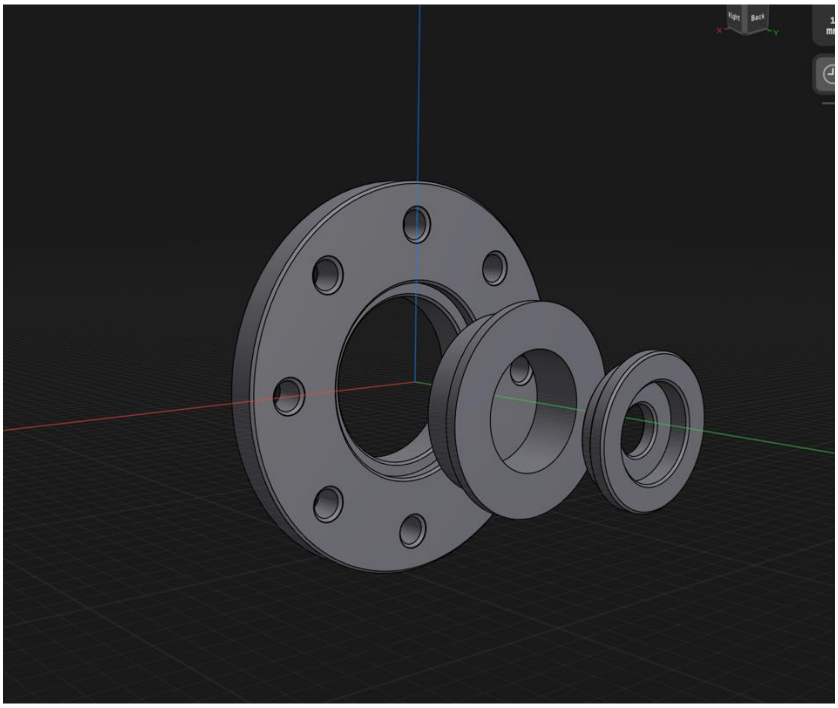

Figure 10: CAD of Horn and Bearing Set

Figure 11: 3D Printed Horn and Bearing Set

3.3.2.2 Torque and Precision Control

One of the main reasons the MX-64 motors were selected for this project was their ability to provide high torque in a compact form factor. Each MX-64 motor is capable of producing a stall torque of up to 6.0Nm at 12V, which was crucial for supporting the weight of the robot arm segments and allowing precise movements in the pitch and yaw axes (see 3.3.3). This torque capability allowed the robot to maintain stability during operation, although some limitations were observed during testing (4.2.2).

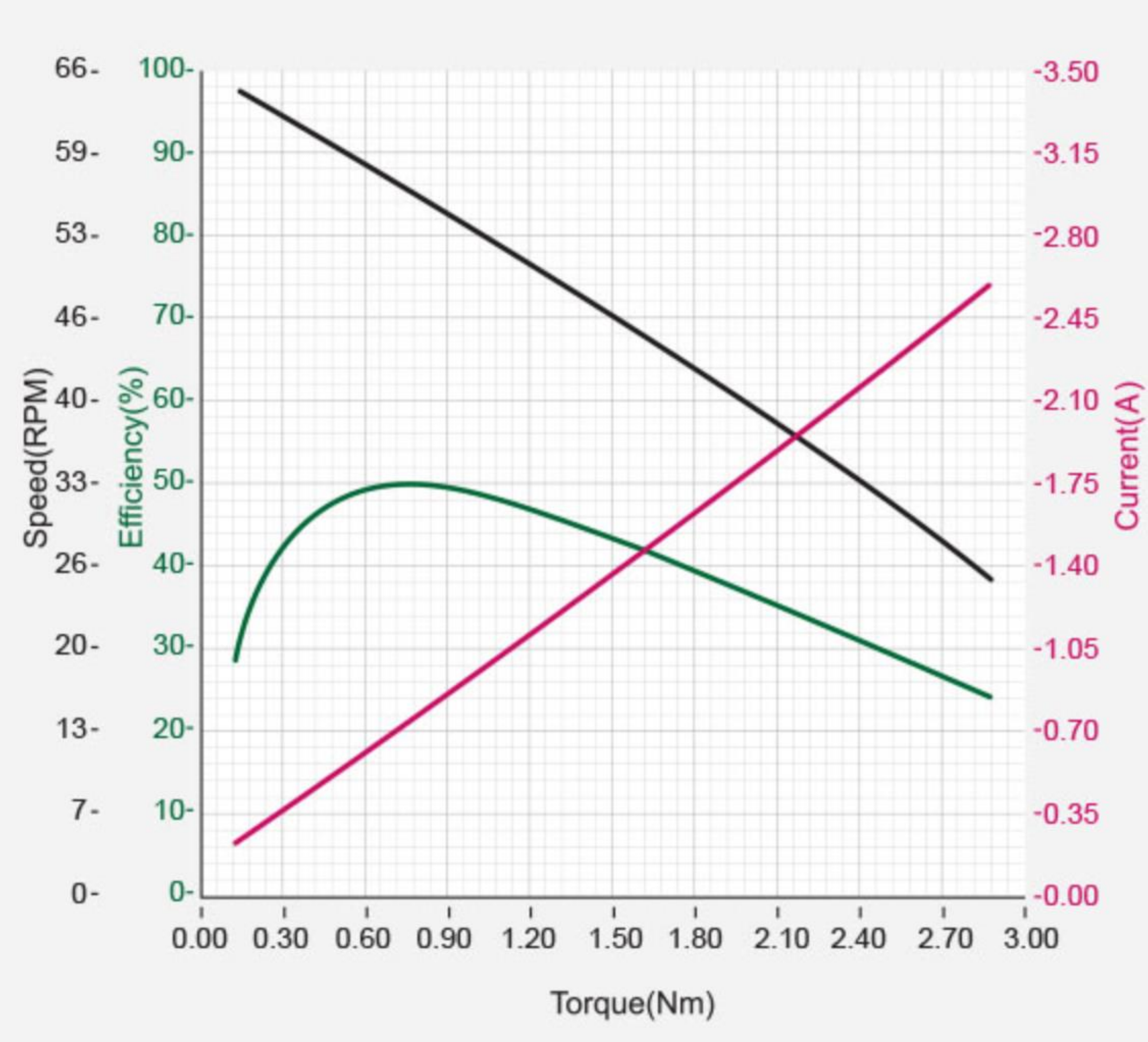

Figure 12: Graph for Torque capabilities of motor w.r.t. Current and Speed[28]

In addition to their torque capabilities, the precision control offered by the MX-64 motors was another critical factor. The motors are equipped with a 12-bit resolution encoder, which provides detailed feedback on the motor's position, speed and load. This high-resolution feedback was essential for achieving accurate control of the robot's joints, allowing for smooth, coordinated movements necessary for tasks such as object manipulation or forming complex shapes like coils.

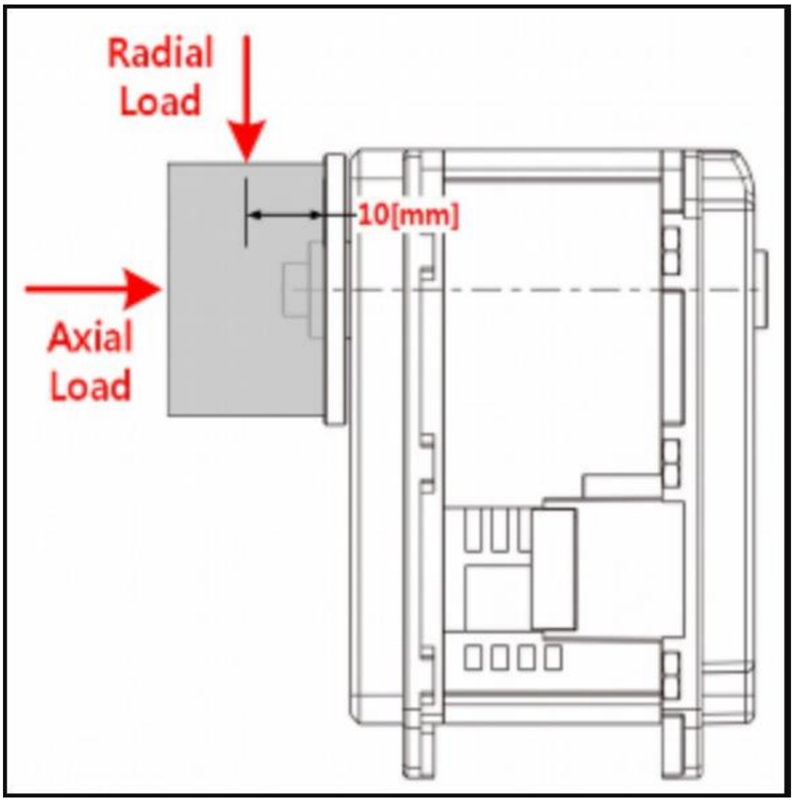

Figure 13: Diagram for Load applied on the motor[28]

3.3.2.3 Additional Features Enhancing Performance

The MX-64 motors come with several additional advanced features that further enhanced their suitability for this project:

- **Programmable Control Modes**: The motors support multiple control modes, including position control, torque control and extended position control. This versatility allowed for a range of experiments to determine the best settings for the robot's tasks. For example, during the Coil Formation Test (4.2.2.2), the torque control mode was particularly useful in managing the forces applied on the robot's joints while attempting to forma a coil shape, thereby improving the robot's adaptability in different operational scenarios.

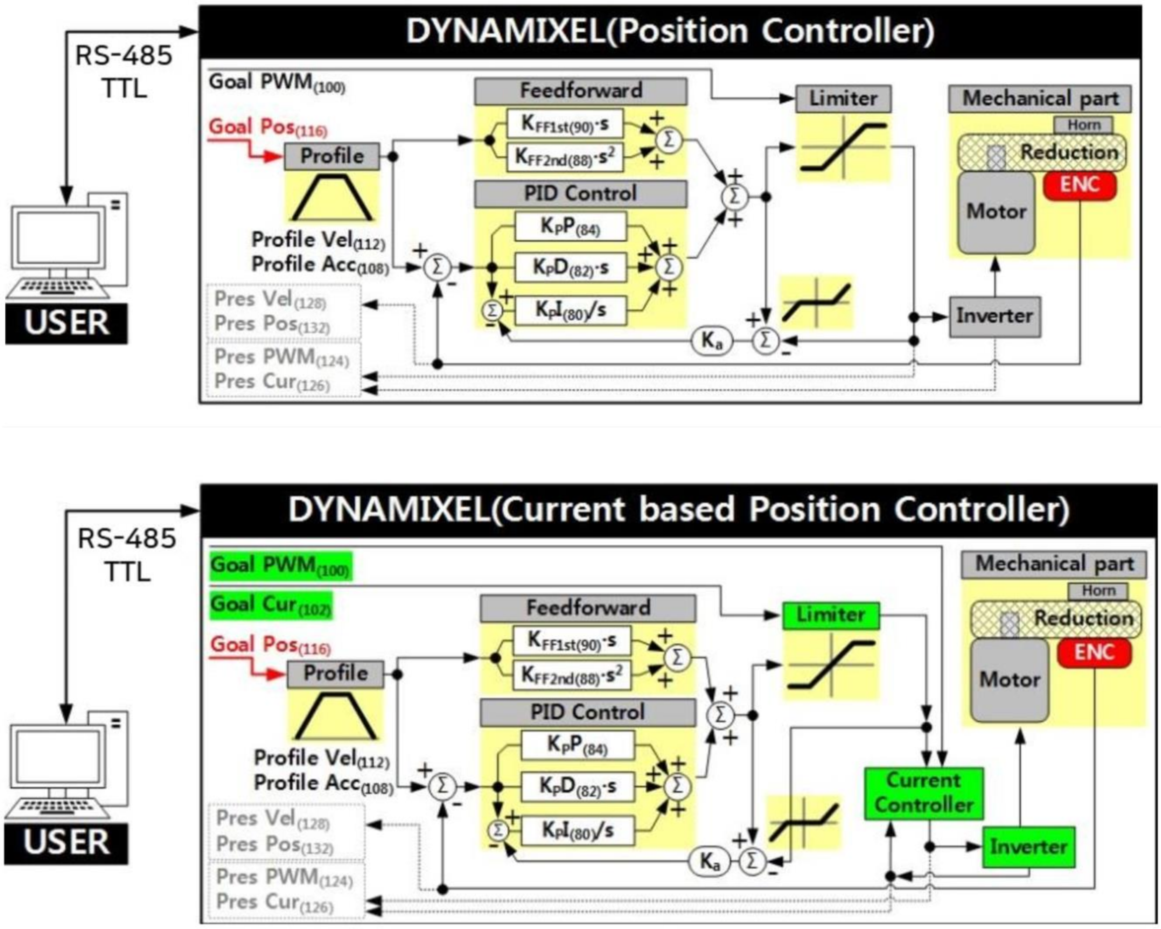

Figure 14: Communication in control modes available to MX-64 Motors

- **Built in Failsafe Mechanisms**: The MX-64 feature built-in safety mechanisms that automatically trigger in the event of overheating, overcurrent or excessive torque. These safety features were crucial for preventing motor damage during rigorous testing, especially when the robot was subjected to loads that exceeded the motors' limits. For instance, in both test cases, the failsafe mechanisms prevented damage when the pitch motors exceeded their torque limits, demonstrating the importance of these safety features in protecting the equipment.
- **Daisy-Chaining Capabilities**: The motors support daisy-chaining, allowing multiple motors to be in series with a single communication line. This capability significantly simplified the wiring and communication setup of the robot arm, reducing the overall complexity of the electrical assembly (this is detailed in 4.1.2).

Additional documentation for the motors can be found on the datasheet for them online [28].

3.3.3 Simplified Kinematics of the Robot Arm

3.3.3.1 Simplifying Approach to Kinematics

Kinematic refers to the study of motion without considering the forces that cause it. In robotics, this normally includes calculating the position and orientation of the robot's end-effector based on joint parameters (referred to as Forward Kinematics) or determining the joint parameters required to achieve a goal effector end-effector position (referred to as Inverse Kinematics). However, given the specific objectives and constraints of this project, a simplified approach was used.

The tentacle arm is designed with multiple segments connected by joints, each of which is actuated by the Dynamixel MX-64 motors. These motors provide two degrees of freedom (DOF) at each joint by controlling rotations around pitch and yaw axes. The pitch axis allows the joint to move up and down, while the yaw axis enables side-to-side rotations. This configuration allows a wide range of motions and is suited for the manipulation tasks required in this project (Figure 15).

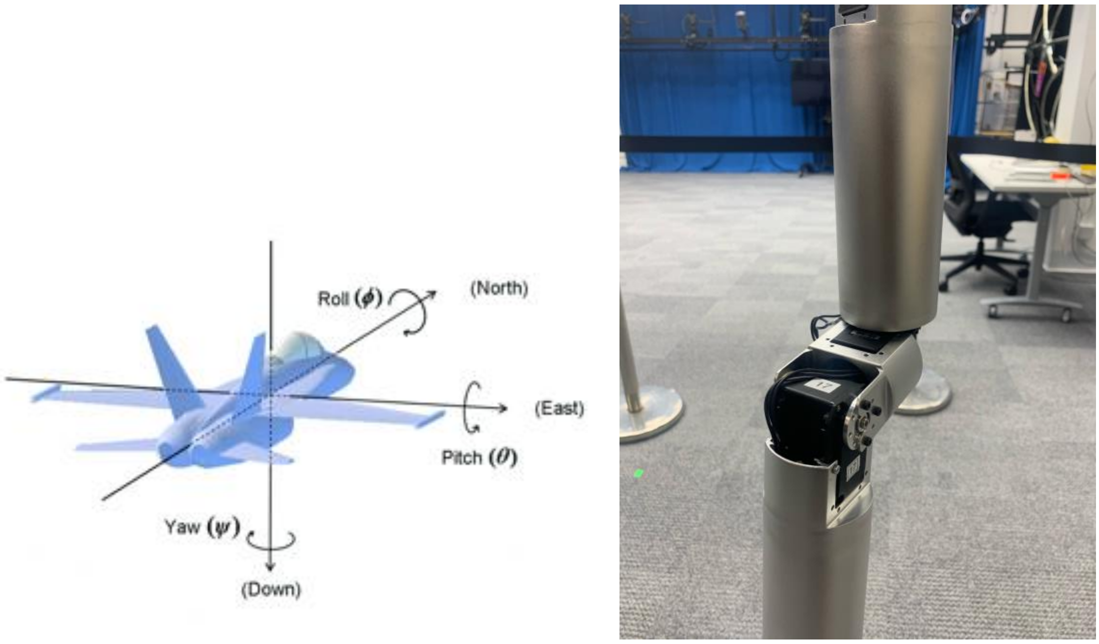

Figure 15: Illustration of Pitch and Yaw Joint Setup[29]

The main concern in this simplified kinematic approach was ensuring that each actuator had sufficient torque to support the weight of the arm segments and carry out the required movements along these two axes. Key focus areas included:

- Torque Requirements at each joint: The torque needed at each joint was calculated to determine if the MX-64 motors could generate enough force to move the arm and maintain its stability during operation. Given the two degrees of freedom at each joint, the analysis specifically considered the torque required for both pitch and yaw movements, with a priority on the pitch axis, where gravitational forces had the most significant impact. Generally, the formula used in these calculations when only considering the pitch axis was:

  $$\tau = r \times F$$

  Where τ is the torque required, r is the distance from the joint to the centre of mass of the segments.

- Torque Distribution along the arm: The analysis also looked at how the required torque would vary along the length of the arm, focusing on the joints closer to the base that carry more load.

Results from this analysis can be seen in Table 2.

Although, this approach is not a comprehensive kinematic model, it is still under the umbrella of kinematics when considering the relationship between joint movements and the robot's configuration. For this project, the focus was more on the torque requirements to achieve the goal positions and movements, given the robot's two DOF per joint, rather than the precise geometric relationships between all the robot's joints and links.

3.3.3.2 Torque Analysis for Two Degrees of Freedom

For a more advanced look on the kinematics of the robot, the torque required at each joint was estimated based on the following considerations:

- Joint Loads and Degrees of Freedom: The load on each joint was estimated based on the weight of the arm segments and the desired position. For example, presumably, the torque required τ required at a joint to lift an arm L and mass m in a gravitational field can be estimated as:

  $$\tau_{pitch} = m \times g \times \frac{L}{2} \times \cos \theta_{pitch}$$

  Where m is the mass of the segment, g is the acceleration due to gravity, L is the length of the segment and θ is the angle of elevation of the arm segment from the vertical along the pitch axis.

While the above equation was estimated, an accurate reading of the angles with respect to the vertical could not be correctly justified in the test scenarios, therefore it was decided to exclude any results from this equation. An approximation of these calculations can be found in the appendix (Table 3).

3.4 Control Systems

The control systems for the tentacle robot arm were designed for manual operation and flexible customisation, which could enable and operator to perform precise manipulation tasks. Two primary methods were used to control the robot: the Dynamixel Wizard UI and a custom Python script developed using the Dynamixel SDK.

3.4.1 Dynamixel Wizard UI

The Dynamixel Wizard UI is a user-friendly interface that is provided by Robotis, which is specifically designed for managing and operating Dynamixel actuators, such as the MX-64 motors used in this project. With this Graphical User Interface (GUI), the operator is able to easily configure, monitor, and control the actuators in real-time. Some of the key features of this UI include:

- **Performance Visualisation**: One of the major advantages of using the UI was its ability to present actuator performance data easily and clearly to the operator. For example, were an actuator to exceed its torque limit and overload, the user could easily locate which actuator had been overloaded and easily reboot it. This feature was particularly useful during the testing phase of the project, where real time adjustments and monitoring were crucial.
- **Ease of Use**: The UI is designed for operators to easily understand how to adjust parameters and execute commands through a simple interface. Furthermore, plenty of documentation can be found online to help new users understand how to effectively use the UI [30].
- **Real-Time Monitoring**: The interface displays information about each actuator connected, including position, baud rate, temperature, load and voltage. This feedback is crucial for monitoring the performance of the MX-64 actuators.

Given its ease of use and real-time monitoring capabilities, the UI was the preferred method for controlling the robot arm during testing. The ability to clearly see information about each actuator clearly on the screen reduced the likelihood of errors and therefore enhanced overall safety.

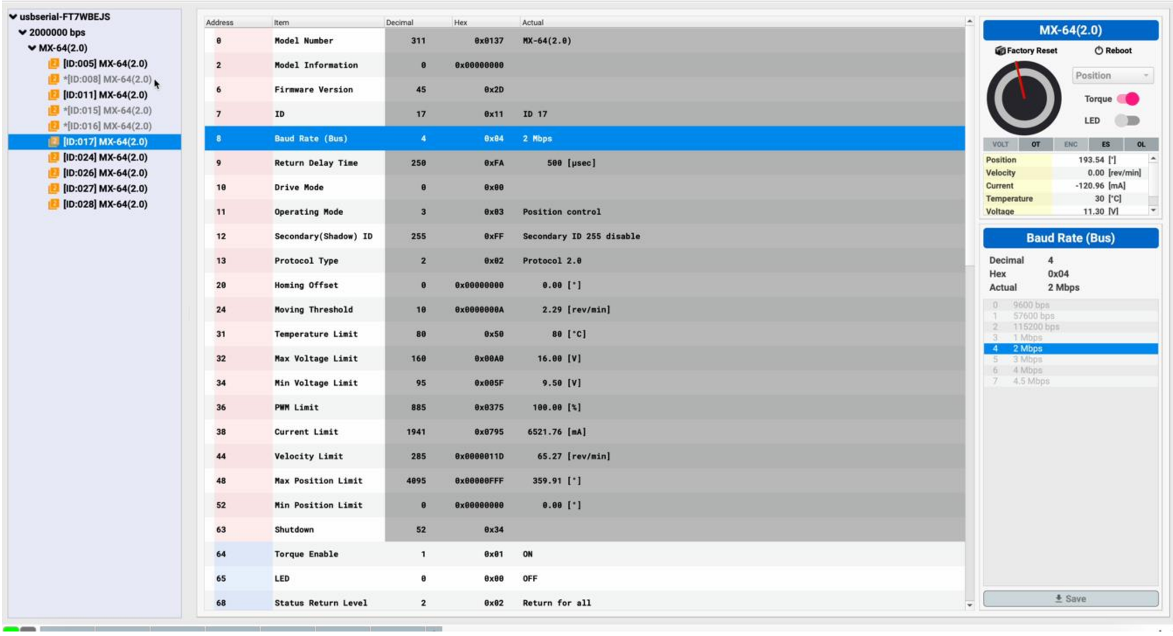

Figure 16: Dynamixel Wizard UI

3.4.2 Python Script

While the Dynamixel Wizard UI provided a strong solution for manual control, a custom Python script was developed using the Dynamixel SDK to offer greater flexibility and potential automation capabilities. The Dynamixel SDK is a software development kit (also provided by Robotis) that allows developers to create custom control programs for Dynamixel actuators. The SDK supports other programming languages such as C++ and MATLAB, but Python was picked due to its ease of use and potential in development for automation.

The creation Python script allowed the following potential goals to be accomplished:

- **Automated control**: Unlike complete manual control offered by the Dynamixel UI, the Python script allowed for automated sequences of movements.
- **Customised Commands**: Precise commands could be sent to the actuators, such as position settings, velocity control and torque adjustments.
- **Data Logging**: Actuator performance could be programmed to be sent back to the operator. This data would be useful for making improvements to the robot's control algorithms.

The Python script provided a potential solution to enabling automation for the tentacle arm in this project. However, due to the risk of damaging additional inventory (specifically the actuators) while performing tests, it was decided that the script would primarily be used for manual control of robot only. Subsequently, since manual control was prioritised, the Dynamixel Wizard UI became the obvious best choice for control and testing.

```python
# Dynamixel IDs
DXL_IDS = [8, 28, 27, 26, 25, 24, 11, 16, 17, 15, 5]

DEVICENAME = '/dev/tty.usbserial-FT7WBEJS'

TORQUE_ENABLE = 1
TORQUE_DISABLE = 0
DXL_MOVING_STATUS_THRESHOLD = 20

# Initialize PortHandler instance
portHandler = PortHandler(DEVICENAME)
packetHandler = PacketHandler(PROTOCOL_VERSION)

if portHandler.openPort():
    print("Succeeded to open the port")
else:
    print("Failed to open the port")
    print("Press any key to terminate...")
    getch()
    quit()

if portHandler.setBaudRate(BAUDRATE):
    print("Succeeded to change the baudrate")
else:
    print("Failed to change the baudrate")
    print("Press any key to terminate...")
    getch()
    quit()

# Enable torque for all motors
for DXL_ID in DXL_IDS:
    dxl_comm_result, dxl_error = packetHandler.write1ByteTxRx(portHandler, DXL_ID, ADDR_TORQUE_ENABLE, TORQUE_ENABLE)
    if dxl_comm_result != COMM_SUCCESS:
        print("%s" % packetHandler.getTxRxResult(dxl_comm_result))
    elif dxl_error != 0:
        print("%s" % packetHandler.getRxPacketError(dxl_error))
    else:
        print("Dynamixel#%d has been successfully connected" % DXL_ID)

def main_menu():
    while True:
        print("\nMain Menu:")
        print("Select a motor to control (Enter ID number) or press ESC to quit:")
        print("Available Motors:", DXL_IDS)
        user_input = getch()

        if user_input == chr(0x1b):  # ESC key
            return None
        elif user_input.isdigit() and int(user_input) in DXL_IDS:
            return int(user_input)
        else:
```

```python
def control_motor(dxl_id):
    print(f"\nControlling Dynamixel#{dxl_id}. Use arrow keys to control. Press SPACE to return to main menu.")
    current_position = (DXL_MINIMUM_POSITION_VALUE + DXL_MAXIMUM_POSITION_VALUE) // 2

    while True:
        print(f"Current Position: {current_position}")
        key = getch()

        if key == ' ':  # Space bar to return to main menu
            break
        elif key == '\x1b[C':  # Right arrow key
            current_position = min(DXL_MAXIMUM_POSITION_VALUE, current_position + 10)
        elif key == '\x1b[D':  # Left arrow key
            current_position = max(DXL_MINIMUM_POSITION_VALUE, current_position - 10)

        # Write goal position
        param_goal_position = [DXL_LOBYTE(DXL_LOWORD(current_position)), DXL_HIBYTE(DXL_LOWORD(current_position)),
                               DXL_LOBYTE(DXL_HIWORD(current_position)), DXL_HIBYTE(DXL_HIWORD(current_position))]
        dxl_comm_result, dxl_error = packetHandler.write4ByteTxRx(portHandler, dxl_id, ADDR_GOAL_POSITION, current_position)
        if dxl_comm_result != COMM_SUCCESS:
            print("%s" % packetHandler.getTxRxResult(dxl_comm_result))
        elif dxl_error != 0:
            print("%s" % packetHandler.getRxPacketError(dxl_error))

# Main loop
while True:
    selected_motor = main_menu()
    if selected_motor is None:
        break
    control_motor(selected_motor)

# Disable torque for all motors
for DXL_ID in DXL_IDS:
    dxl_comm_result, dxl_error = packetHandler.write1ByteTxRx(portHandler, DXL_ID, ADDR_TORQUE_ENABLE, TORQUE_DISABLE)
    if dxl_comm_result != COMM_SUCCESS:
        print("%s" % packetHandler.getTxRxResult(dxl_comm_result))
    elif dxl_error != 0:
        print("%s" % packetHandler.getRxPacketError(dxl_error))

portHandler.closePort()
print("Program terminated.")
```

Figure 17: Python Script for Manual Control

4 ASSEMBLY, TESTING AND APPLICATIONS OF THE TENTACLE ROBOT

4.1 Assembly Process and Integration

4.1.1 Mechanical Assembly

The mechanical assembly of the tentacle robot arm was carefully designed following the considering testing in the RViz simulation to optimise its ability to grasp objects. The ceiling was selected as the most suitable base for mounting the robot not only because it provided the most mechanical stability for robot operation, but also as a top-down control approach leverages gravity to enhance the robot's performance and provided the most optimal efficiency for the conceptualised test environment for its application in space. This provided the best choice to simulate the arm's potential in space application, were the robot set up on the ground or the wall, the segments would combat the force of gravity on the entire robot itself thereby decreasing manueverability as the actuators would not be able to generate enough torque to sustain the weight of the robot.

The arm segments are constructed from aluminium, with hollow ends designed to accommodate the actuators. Each actuator is secured in place using standard M2.5 screws, which provide robust attachment and precise control over the movement of each segment. At each joint (excluding the base and the end of the robot), the actuators are doubled, using an additional aluminium joint structure. These joints serve both as connectors between different arm segments and as a mechanism to control each segment's movement along the pitch and yaw axes once the robot is extended from the ceiling, thereby allowing multiple degrees of freedom (Figure 20).

The arm segments are arranged in a "stair-like" configuration (Figure 19), which, as determined from previous simulations, offers the best operational performance. This arrangement increases the robot's flexibility and allows it the potential to wrap around objects, assuming gravity is not a limiting factor.

Figure 18: Suboptimal link between arm segments reducing flexibility.

Figure 19: "Stair-like" connection between arm segments

Figure 20: Sample Joint Assembly shown with Actuator alignment. These structures are the primary points of control for the robot.

4.1.2 Electrical Assembly

In addition to the mechanical assembly, establishing the electrical connections and ensuring reliable communication between the actuators and the user control system were important components of the assembly process. These electrical connections were essential for supplying power to each motor and allowing seamless control of the robot arm.

As stated in the mechanical assembly section (4.1.1), the aluminium arm segments have hollow ends to accommodate the actuators. The robot is powered using a standard DC power connector port, which is connected to a U2D2 Power Hub Board (Figure 21). This board, combined with the U2D2- a compact USB communication converter (Figure 22)-serves as both the power distribution center and the control hub for operating the Dynamixel equipment from a computer.

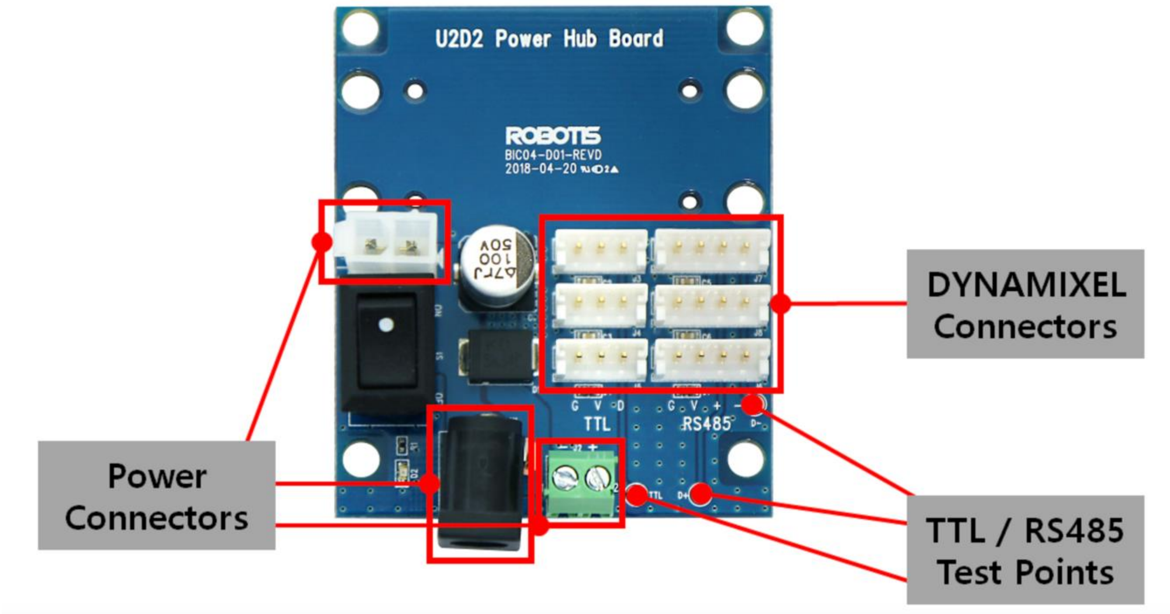

Figure 21: U2D2 PHB Layout[31]

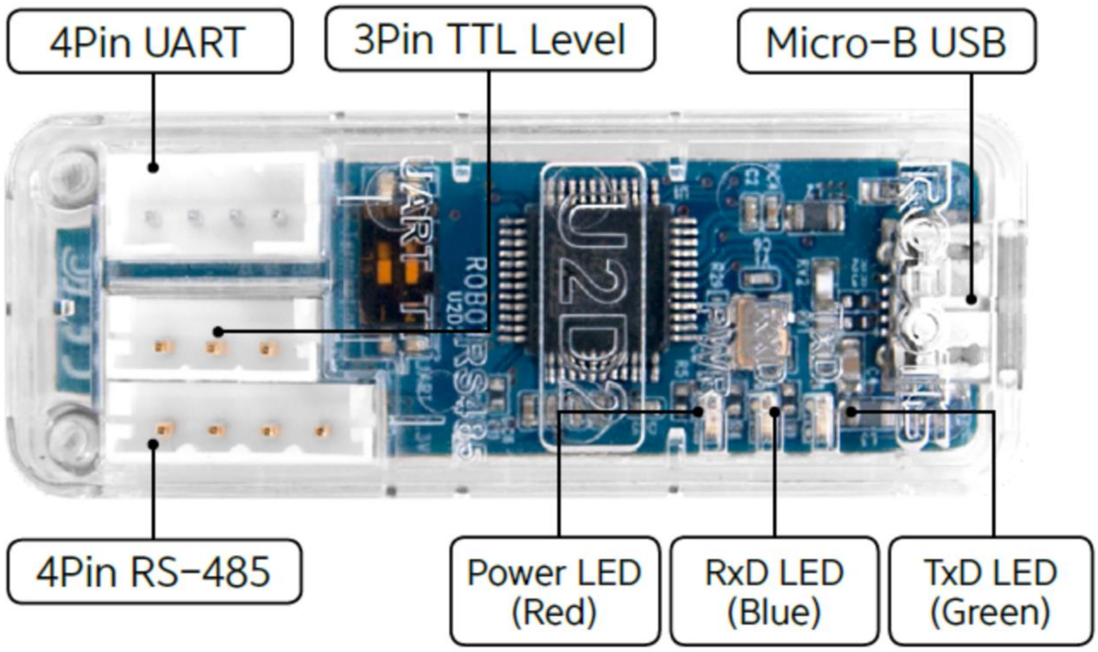

Figure 22: U2D2 Layout[32]

The U2D2 PHB distributed power from a DC source to the Dynamixel MX-64 actuators through a series of 3-pin TTL connector ports [31]. This board is essential for providing consistent power delivery to each motor while maintaining a central point for all electrical connections (Figure 23).

The U2D2 USB communication converter is responsible for converting USB signals from a computer into TTL signals that the actuators can understand [32]. It acts as a bridge between the computer and the robot's actuators, allowing for real-time control and feedback. The computer sends commands through the USB port, which are then converted into control signals by the U2D2, directing the actuators to perform specific movements. Feedback from the actuators, such as position and torque, the two most important data packages from the actuators, is also sent back through the U2D2 to the computer, allowing the user to make precise adjustments while also monitoring the robot's performance.

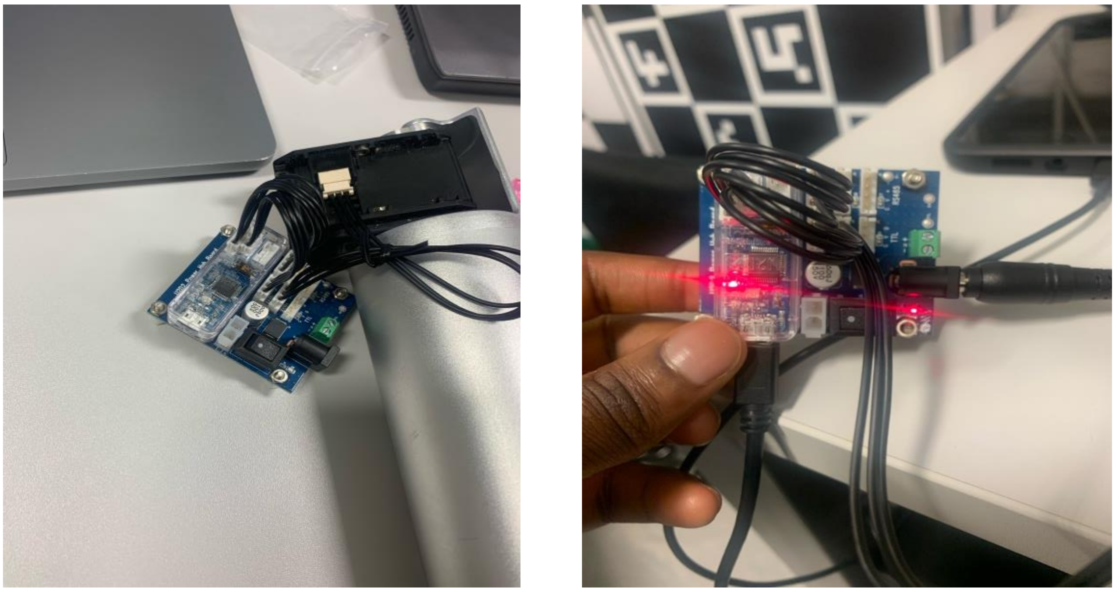

Figure 23: Left: U2D2 PHB Connection to Base. Right: Powered up U2D2 PHB and connected U2D2

A 3-pin TTL cable was used to connect the U2D2 PHB to the base motor (Figure 23). From there, additional 3- pin TTL cables were routed along the arm's length, hidden inside the hollow segments, to connect each motor. During assembly, it was discovered that the inventory cables ordered were slightly shorter than required to reach along the entire length of the arm. To resolve this issue, additional soldering was carried out to extend the cables to the necessary length (Figure 25). Shorter cables were strategically used for connections at the joint structures linking each arm segment to prevent tangling or damage while the robot was in operation.

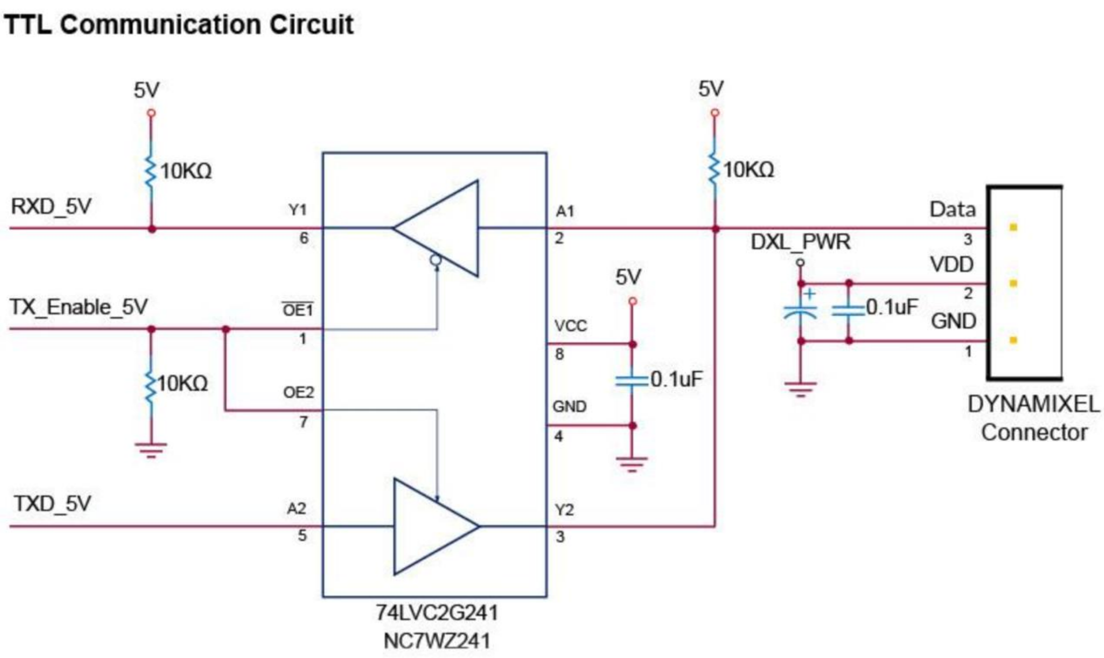

Figure 24: TTL Communication Circuit[28]

Figure 25: Cable modifications made to allow connections

4.2 Mounting, Testing and Case Studies

4.2.1 Safety Concerns and Risk Assessment for Testing

Before beginning the testing phase, several safety concerns were identified and addressed to ensure the safety of both the operators and the equipment. The following measures were implemented:

- **Structural Stability**: The iron rig used to suspend the robot was thoroughly inspected to ensure it could support the full weight of the robot arm, which is approximately 3.456 kilograms. An additional safety cable was passed around the mount of the arm to prevent any accidental damage to the robot during testing.
- **Electrical Safety**: All electrical connections, including the U2D2 PHB and the 3-pin TTL cables, were double-checked for proper insulation and secure fittings. The power was regulated to avoid overvoltage conditions that could damage the motors.
- **Manual Control Precautions**: Since the robot was manually controlled during testing, clear rules were established for the operator to follow. The USB extender was used to allow the user to connect to the U2D2 converter from a safe distance
- **Maximum Risk Area Precaution**: The maximum risk area is the area in which the robot could operate and pose a threat to objects or people. Assuming the robot could be extended the straight out horizontally (parallel to the ground), the maximum risk area was calculated to be approximately 12.57 square meters, given the fact that the approximate full length of the robot was 1.8 meters (rounded up to 2 meters) (Figure 26).

$$A = \pi r^2$$

$$where\ r\ is\ the\ length\ of\ the\ robot\ and\ A\ is\ the\ therotical\ maximum\ risk\ area.$$

Figure 26:Operational Area of the Robot Arm

- **Emergency Stop Protocol**: An emergency stop mechanism was integrated into the power system, allowing the operator to immediately halt the robot's movement by cutting off all power being administered to the robot in case of unexpected behaviour or safety concerns.

4.2.2 Testing and Case Studies

After confirming the proper electrical communication between the motors, the final assembly was completed by attaching each motor and arm segment together. The robot arm was then suspended from an iron rig attached to the ceiling using the frame setup inherited from a previous project (Table 1). The initial configuration of the robot included six arm segments and 11 motors, resulting in total weight of approximately 3.456 kilograms (3456 grams) and an overall length of about 1.8 meters (180 centimetres).

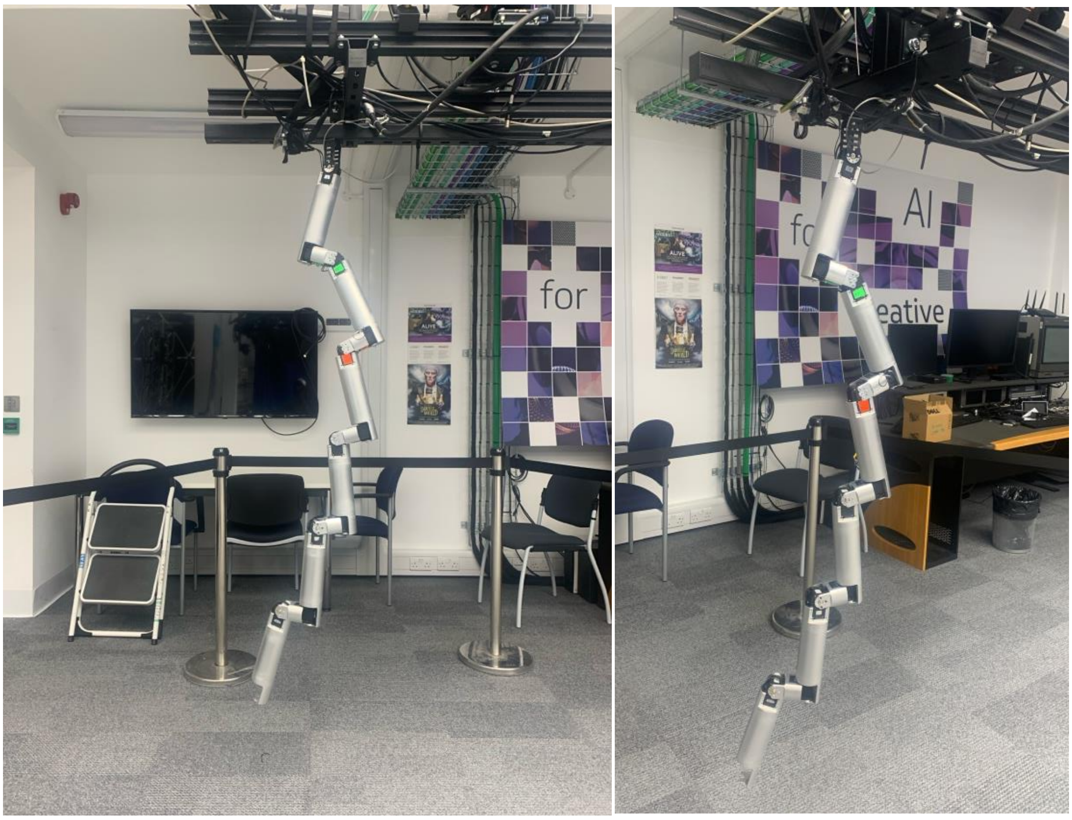

Figure 27: Complete Assembly of Tentacle Robot Arm

To validate the robot's design and functionality as a manipulator in space-like conditions, two test cases were conducted:

- **Case 1: Object Interaction Test**
  In this test, the robot arm was required to manoeuvre and interact with a non-static object, simulating the dynamic nature of objects in space. A successful outcome was defined as the robot arm being able to make contact with the object and demonstrating the ability to manoeuvre around it. This test aimed to assess the robot's agility and precision in tracking and interacting with targets.
- **Case 2: Coil Formation Test**
  The goal of this test was for the robot to successfully form a coil shape. Achieving this shape would demonstrate that the robot meets its design criteria as tentacle-like, snake-like manipulator. Additionally, this capability is important for potential applications in asteroid mining, where the robot may need to wrap around irregularly shaped objects to manipulate or extract resources.

It is important to note that in both test cases, the robot was manually controlled by a user. As a result, the outcomes were significantly influenced by the user's decision making and the specific joints activated to control the robot. This factor introduces variability in the results, but successful execution of the tests would still demonstrate the robot's capabilities and its potential for space applications.

4.2.2.1 Case 1: Object Interaction Test

- **Objective**: To evaluate the robot's ability to interact with a non-static object.
- **Setup**

  The robot was configured in its default setup as described in the testing section (4.2.2). The table below details the control specifications for each actuator during this test, with moto locations corresponding to image (Figure 27) from top to bottom. The table highlights the torque capabilities of all motors responsible for pitch control within a range of +/- 90 degrees. The yaw motor's capabilities have been excluded from the table as, depending on their operating angle, they do not bear additional load and are therefore negligible in this scenario (see 3.3.3.1):

ACTUATOR/MOTOR SPECIFICATIONS (CASE 1)				
Motor ID	Axis Control	Arm Segments Controlled	Approximate Force Applied (Newtons)	Torque Required (Newton metres)
08(Base Motor)	Pitch	6	17.28	5.02
28	Yaw	5		
27	Pitch	5	28.80	8.31
26	Yaw	4		
25	Pitch	4	23.04	6.68
24	Yaw	3		
11	Pitch	3	17.28	5.02
16	Yaw	2		
17	Pitch	2	11.52	3.34

15	Yaw	1		
05	Pitch	1	5.76	1.67

Table 2: Actuator information for Case 1

The non static object used in this case experiment was balloon attached to a mic stand. The mic stand allowed different heights to be simulated allowing more difficult scenarios for the robot to adapt around to be tested. The choice of the balloon was ideal for the concept of an object in space as a balloon will move once interacted with, requiring careful operator handling to attempt to grasp and interact with it.

- **Results & Challenges**

  As shown in Figure 28 below, the robot successfully interacted with the test object. The robot was able to make contact with the balloon and manoeuvre around it, demonstrating its capability to interact with other objects within its operational range. This result confirms that the robot meets the objective of this test, showcasing its potential as a manipulator in space environments.

Further tests were conducted by adjusting the height of the microphone stand to introduce different levels of difficulty for the robot to interact with the non-static object. These additional tests allowed the exploration of the torque limits of the pitch motors, as estimated in Table 2. During these experiments, it was observed that the pitch motors controlling joints 1 to 3 were unable to fully support the weight of the robot's structure, meeting expectations from initial torque calculations. Although the motors generated sufficient generated sufficient power to attempt holding the structure, the MX-64 motors activated their built-in failsafe mechanisms to prevent potential damage from overloading once the torque limit is met.

Another challenge encountered during the tests was the influence of human error. The robot was manually controlled via a computer interface (see 3.4), and it is unclear whether the chosen joint movements or strategies were optimal. This limitation highlights the need for improved control mechanism, which will be discussed in detail later on.

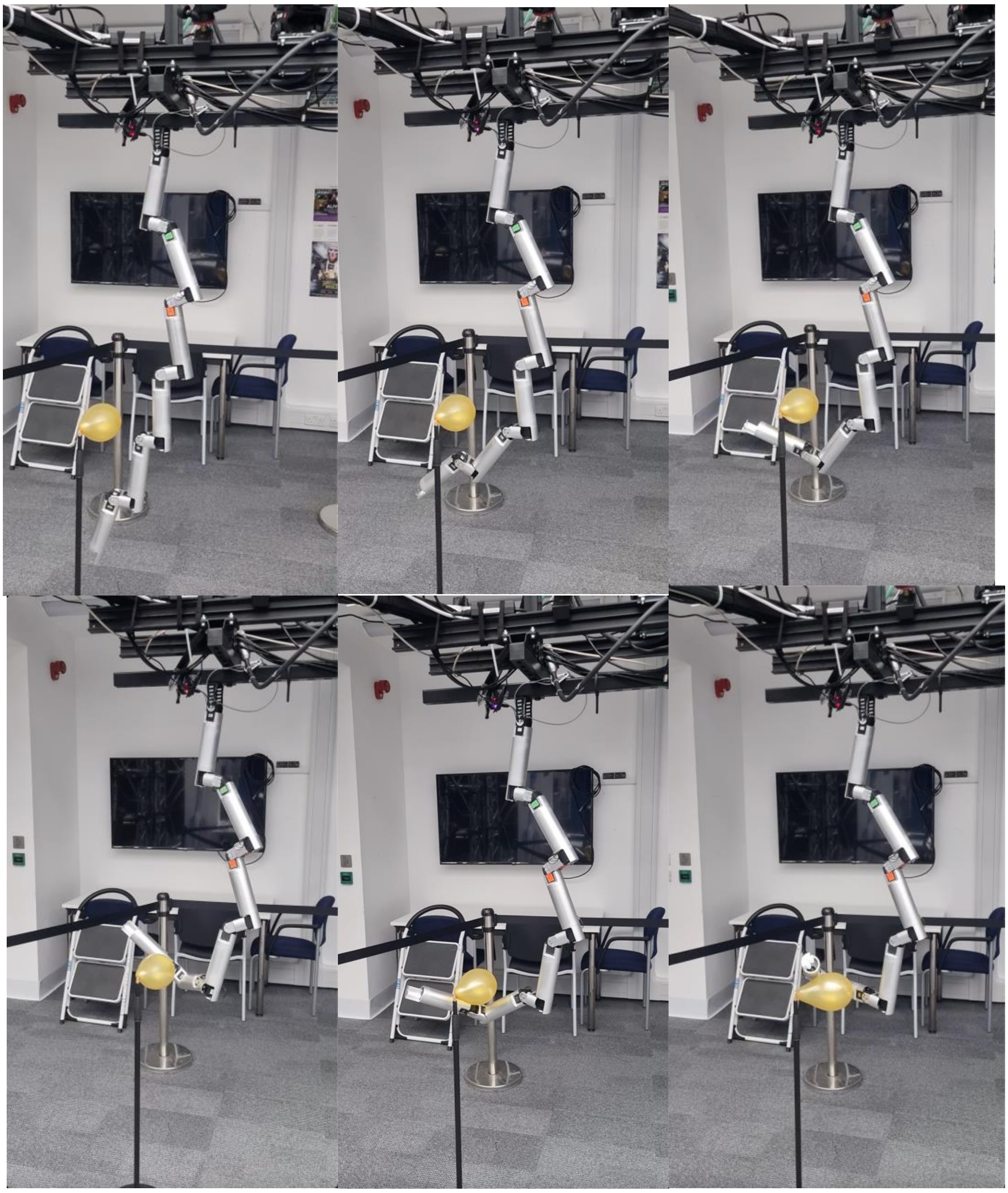

Figure 28: Video Frame Snapshots of Case 1 Test.

4.2.2.2 Case 2: Coil Formation Test

- **Objective**: To demonstrate the robot's ability to form a coil shape and assess its flexibility
- **Setup**

  The robot was initially configured in its default setup, similar to the configuration used in Case 1. However, during the testing phase, modifications were made to reduce the overall weight of the robot. One arm segment was removed, bringing the total weight to 2880 grams. Additionally, elastic rubber bands were strategically attached along the first, second and third arm segments using pins. These rubber bands were intended to provide extra support and generate additional torque when the pitch joints near the top of the robot arm flexed into the coil shape. The testing area remained identical to that used in Case 1. In this setup, no objects were placed to be grasped, as any additional items could interfere with the robot's movement and potentially cause entanglement.

- **Result and Challenges**

Achieving the coil shape proved to be more challenging with the robot's default configuration, as anticipated from the torque calculations presented in in Table 2. The target configuration-a coil shape- required the upper joints to flex at angles that shifted the robot's centre of mass, thereby increasing the load on the joints. Although a rough estimate of the load on the first joint could be calculating using specific equations (see 3.3.3.2), the precise accuracy of these calculations was not prioritised. Instead, the focus was on achieving the chosen coil shape, ass further exploration of these calculations falls outside the scope of this dissertation's objectives and timeframe. These considerations will be discussed in more detail in the conclusion of this report.

As a result of these challenges, modifications were made to the robot structure to improve its ability to achieve the coil shape. One arm segment was removed to decrease the overall weight, and rubber bands were added to the frames of the first, second and third arm segments. The idea was that the rubber bands would provide additional support to the motors at the higher joints, generating extra torque to assist in forming the coil shape. After several initial tests, this modification proved to be effective (Figure 29 and Figure 30).

Figure 29: Modifications to help generate more Torque

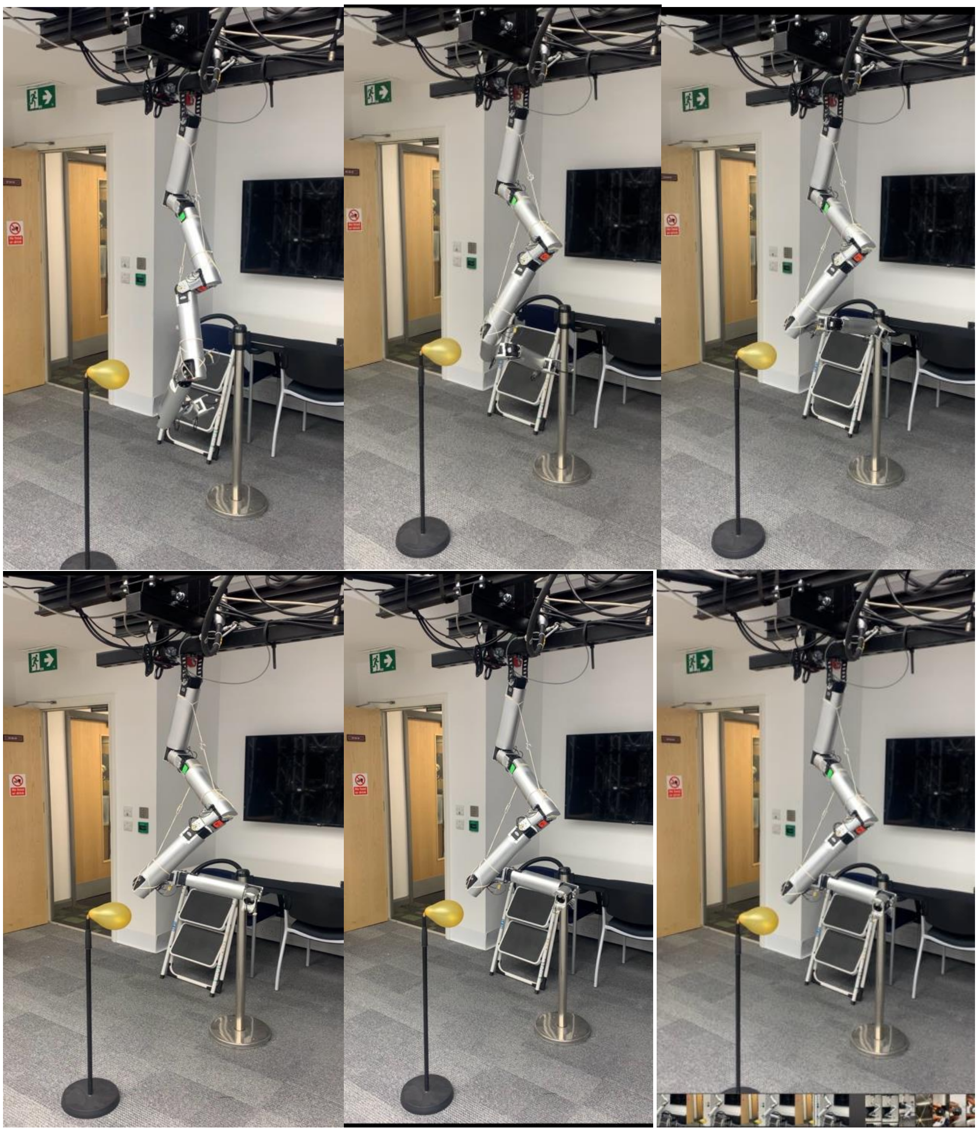

Figure 30: Video frame Snapshots of Case 2

5 CONCLUSION

5.1 Evaluation

In this dissertation, the primary objective was to design, assemble and evaluate a tentacle robotic arm capable of manipulation in space-like conditions, particularly focusing on its application in asteroid mining. The objective also included assessing and assembling the available robotic components, the development of a control system, and testing the robot's capabilities in controlled experiments.

The tentacle arm was successfully assembled using both inherited and some new components. Despite some challenges with the inherited hardware, such as defective actuators, the assembly was completed by using innovative solution like 3D-printed components, demonstrating adaptability and problem-solving skills.

A robust control system was designed using both the Dynamixel Wizard UI and a custom Python script. This allowed the robot to be manually controlled and created a potential avenue for automated control for the robot.

Two test cases were conducted to evaluate the robot's capabilities. In the first test case, the object interaction test, the robot's adaptability and precision in manipulating and interacting with non-static objects was demonstrated, reflecting its potential for handling dynamic environments in space. In the second test case, the coil formation test, the robot's ability to form complex shapes was successfully demonstrated, showcasing its flexibility and potential application in environments requiring complex manipulation like asteroid mining.

5.2 Future Work

Given the potential scale of the project, there are multiple future studies that could done to improve the robot created.

5.2.1 Selection of More Optimal Materials

A potential area for future research is the use of different materials for the robot's joints. The current design uses aluminium segments and 3D- printed plastic horns, which, while effective for initial testing, may not provide the durability or the flexibility needed for more demanding applications, such as dep space exploration or continuous use in extreme environments. Future studies could explore using more lightweight metals or some form of advanced composite material to enhance the robot's strength to weight ratios, increase the arm's load-bearing capacity and improve its resistance to environmental challenges like radiation for space-mission like tests. The choice of materials could also impact the robot's kinematic performance, potentially reducing the torque required at each joint and improving the robot's overall flexibility and range of motion.

5.2.2 Development OF Pneumatic or Tendon-Like Joints

Another potential advancement is the development of pneumatic or tendon-like joints to replace (or improve) the existing motor-driven joints. Pneumatic actuators, which use compressed air to generate movement, could offer an increase in the degree of flexibility and control, mimicking the natural movements of biological organism more closely compared to traditional motors. Similarly, tendon-like joints, which use cables to manipulate the arm's segments could provide more precise movements, particularly in unstructured environments. Both these alternative joint designs could significantly enhance the robot's adaptability, making it more suitable for a larger range of tasks, such as asteroid mining or repairs in microgravity environments.

5.2.3 Reinforcement Learning for Autonomous Control

The current control system used in this project is primarily manually operated. Future work could explore the integration of reinforcement learning algorithms to enable the robot to learn and adapt its movement autonomously. Reinforcement learning, which is a type of machine learning algorithm where a stem learns to make decisions by receiving a form of reward for actions, could allow the robot to improve its movements for tasks. For example, this could be especially in helping the robot in learning to grasp objects of different shapes and sizes more efficiently or learning to navigate complex environments autonomously. This would require extensive simulation work, which could be done with ROS environment possibly, followed by real-world testing to validate the algorithms.

5.2.4 Additional Sensors to Utilise ROS Capabilities

To improve the robot's sensing capabilities, future work could involve integrating additional sensors and using the full capabilities of ROS. The current setup is relatively basic in terms of sensory input, primarily only focused on motor feedback. By adding sensors such as LIDAR or cameras, the robot could obtain a better understanding of its surroundings, allowing more advanced navigation, object recognition and manipulation tasks. These sensors could be integrated into the ROS framework to provide real-time environment mapping or data processing, significantly improving the robot's operational capabilities.

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5.2.5 Testing in Different Environments

The current testing was restricted to a controlled indoor environment, which, while useful for the case studies and initial trials, does not fully simulate the conditions the robot would face in actual deployment scenarios such as asteroid mining. Future work could introduce more challenging environments to better understand the robot's limitations and performance under different conditions. Some of these environments include underwater environments to mimic zero-gravity conditions or vacuum chambers to simulate space. Testing in these environments would provide information into the robot's operational durability and control accuracy under different physical challenges.

5.2.6 Advanced Kinematics and Dynamics Study

While the current project used a simplified kinematic model focusing primarily on torque requirements, a more advanced study using both kinematics and dynamics could provide a better understanding of the robot's movement and control capabilities. This might involve developing comprehensive models that consider the robot's inertial properties, its dynamic interactions between its segments and external forces like gravitational pull. Advanced modelling would also allow for better optimisation of control algorithms and hardware configurations, potentially leading to more efficient robot designs.

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APPENDIX

ACTUATOR CALCULATIONS APPROXIMATED FOR CASE 2				
Joint Position (From Top)	Angle θ (degrees)	Approximate Force Applied (Newtons)	Torque Required (Newton Metres)	Weight(Kilo-grams)
1	60	28.24	2.05	2.88
2	70	16.93	1.00	1.728
3	50	11.29	1.00	1.152
4	40	2.65	0.64	0.576
5	N/A	0	0	0

Table 3: Torque Required Approximations

Link to Gitlab to view code files: https://gitlab.surrey.ac.uk/eo00855/masters-project-tentacle-arm/-/tree/main